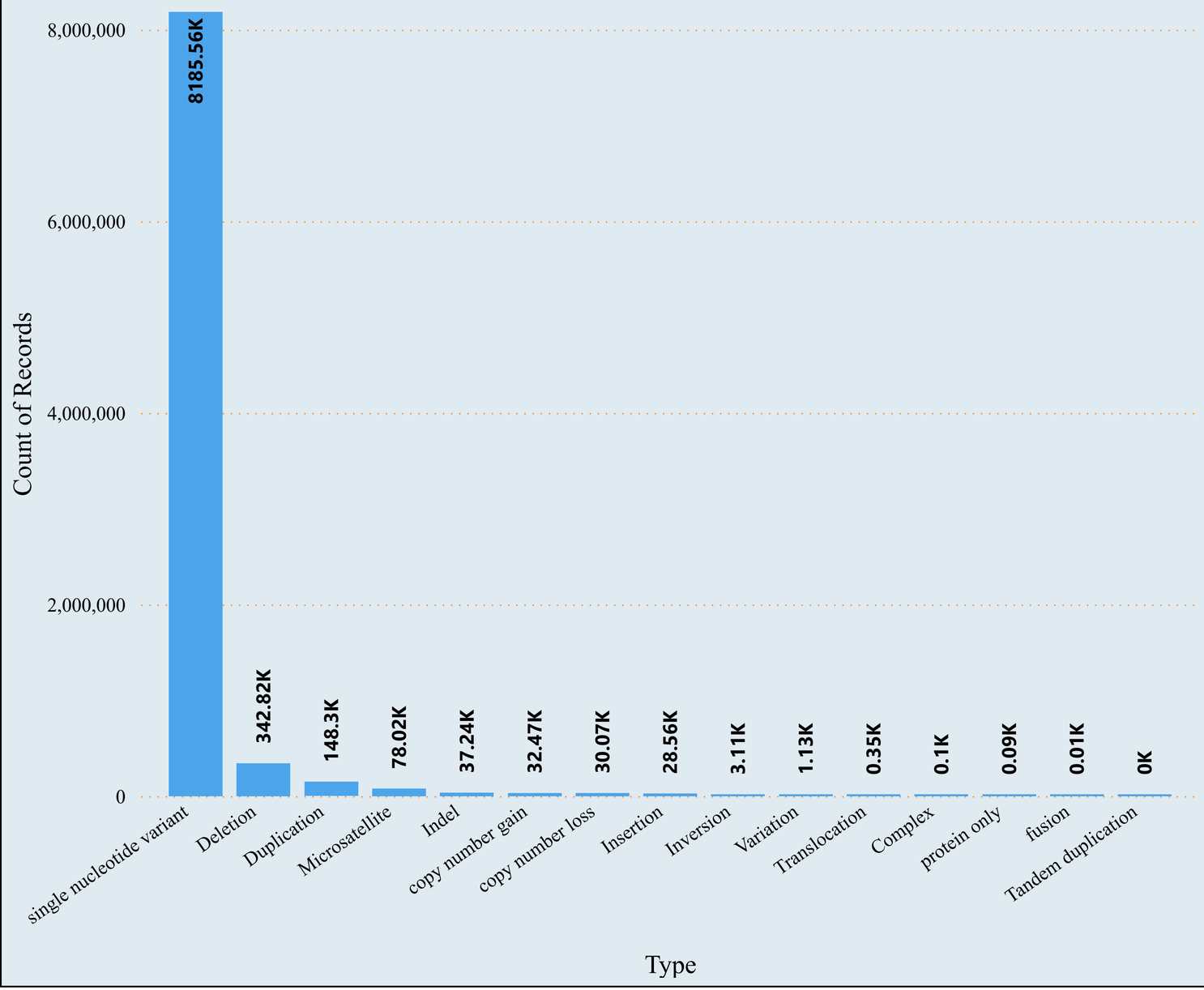
Breakdown of Variant Types & Clinical Significance pt.1

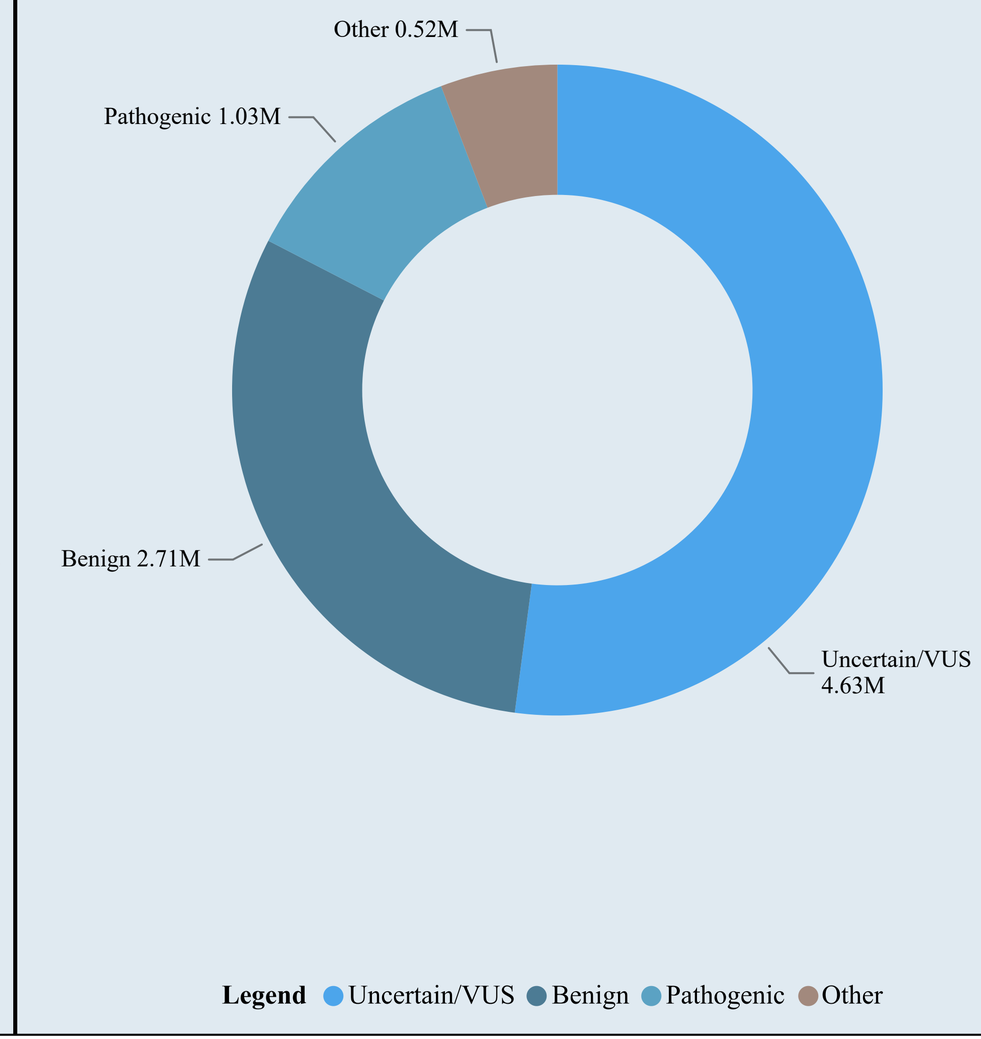
Variant Types

Source: ClinVar variant_summary (n=8.8M records)



Clinical Significance Distribution

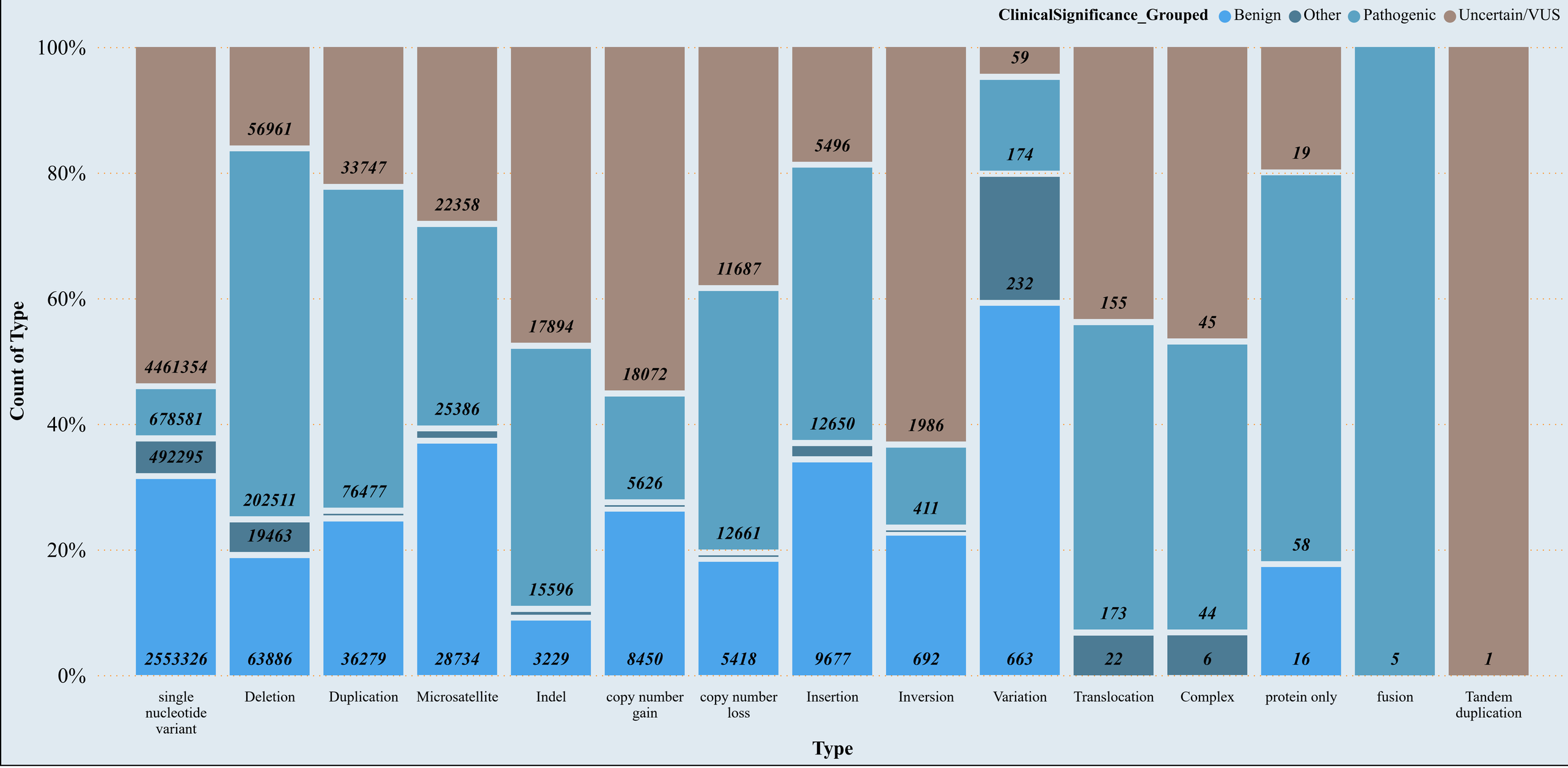
Majority of variants remain clinically uncertain



Breakdown of Variant Types & Clinical Significance pt.2

Clinical Significance by Variant Type

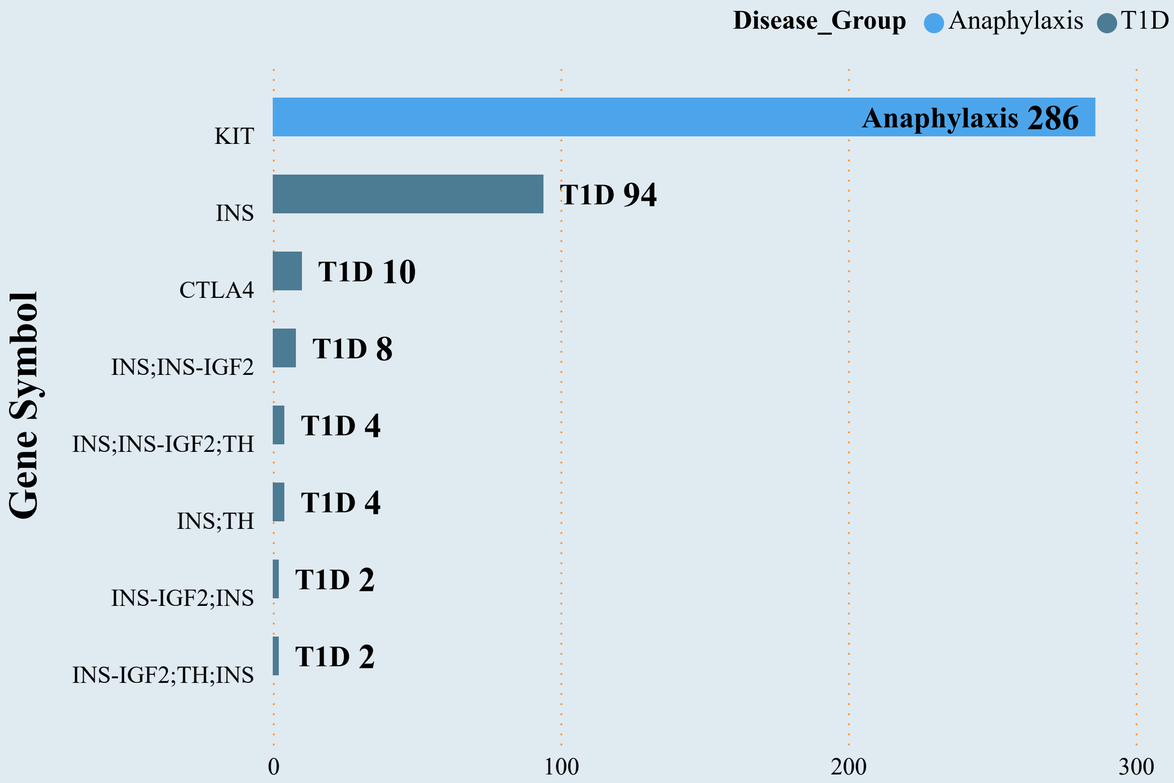
Uncertain/VUS variants dominate across all variant types



T1D vs Anaphylaxis Gene Overlap

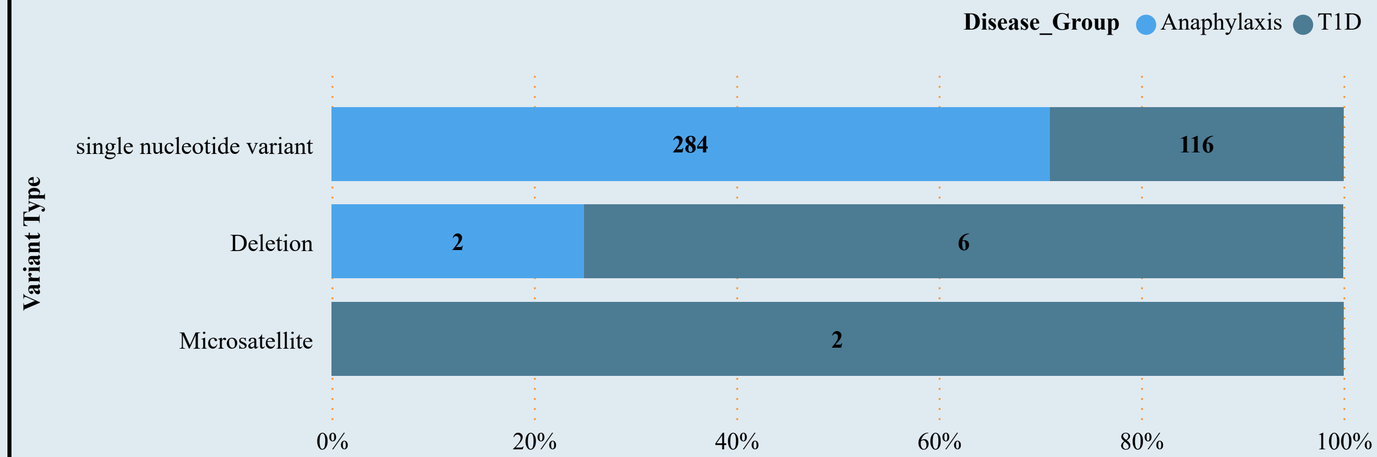
Gene Frequency by Condition Group

No shared genes identified between T1D and Anaphylaxis in ClinVar



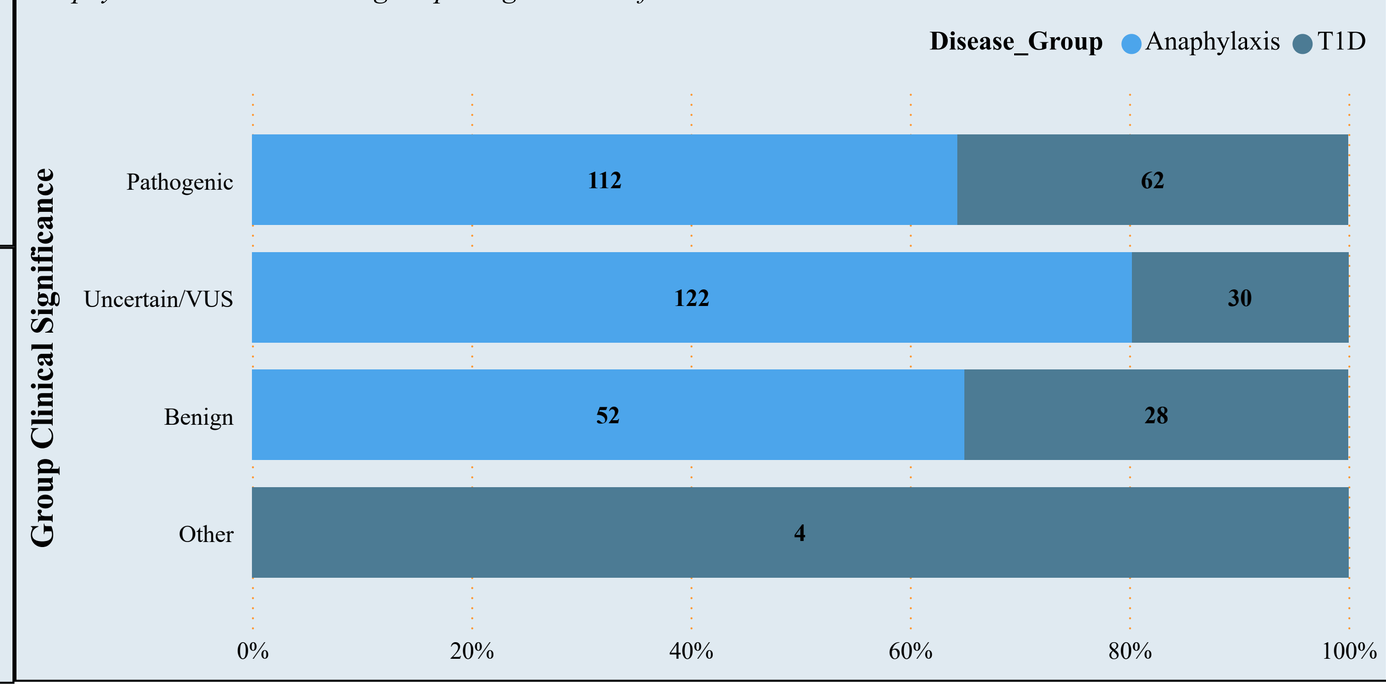
Variant Type by Condition Group

SNVs represent the majority of variants in both disease groups



Clinical Significance by Condition Group

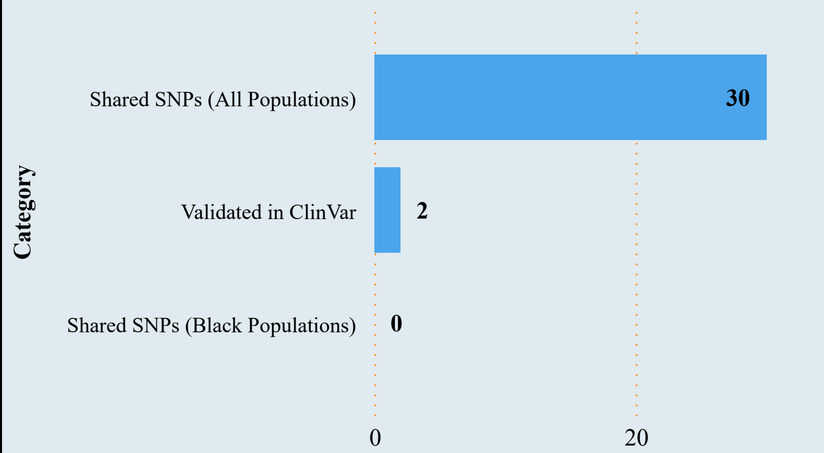
Anaphylaxis variants show higher pathogenic classification than T1D



Anaphylaxis	*T1D*
286	124

Representation Gap in Shared T1D-Allergy Loci

Zero shared loci have been studied in Black populations

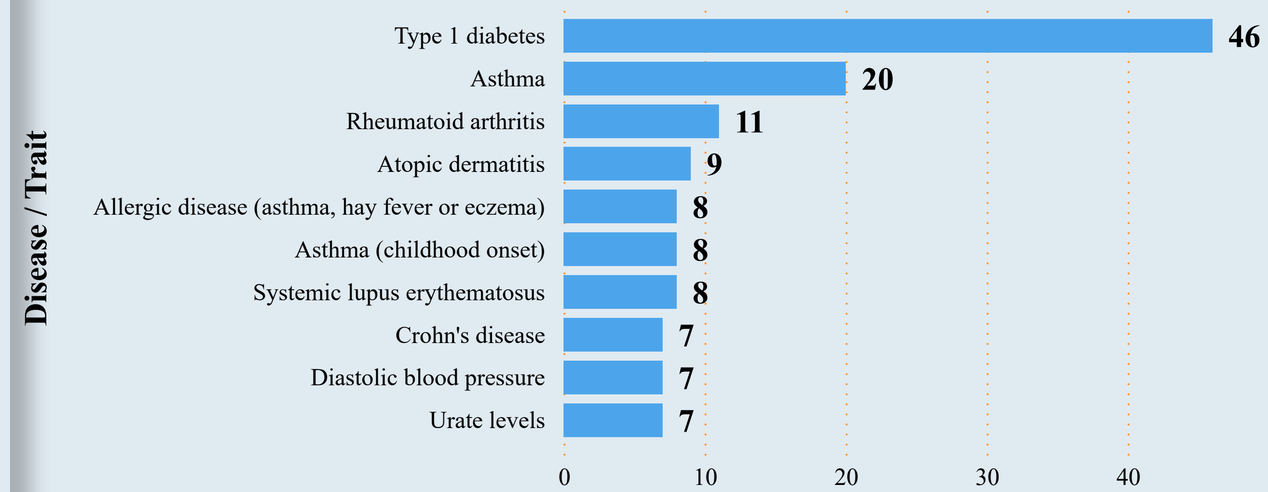


Unique Diseases in GWAS Catalog

5.8K

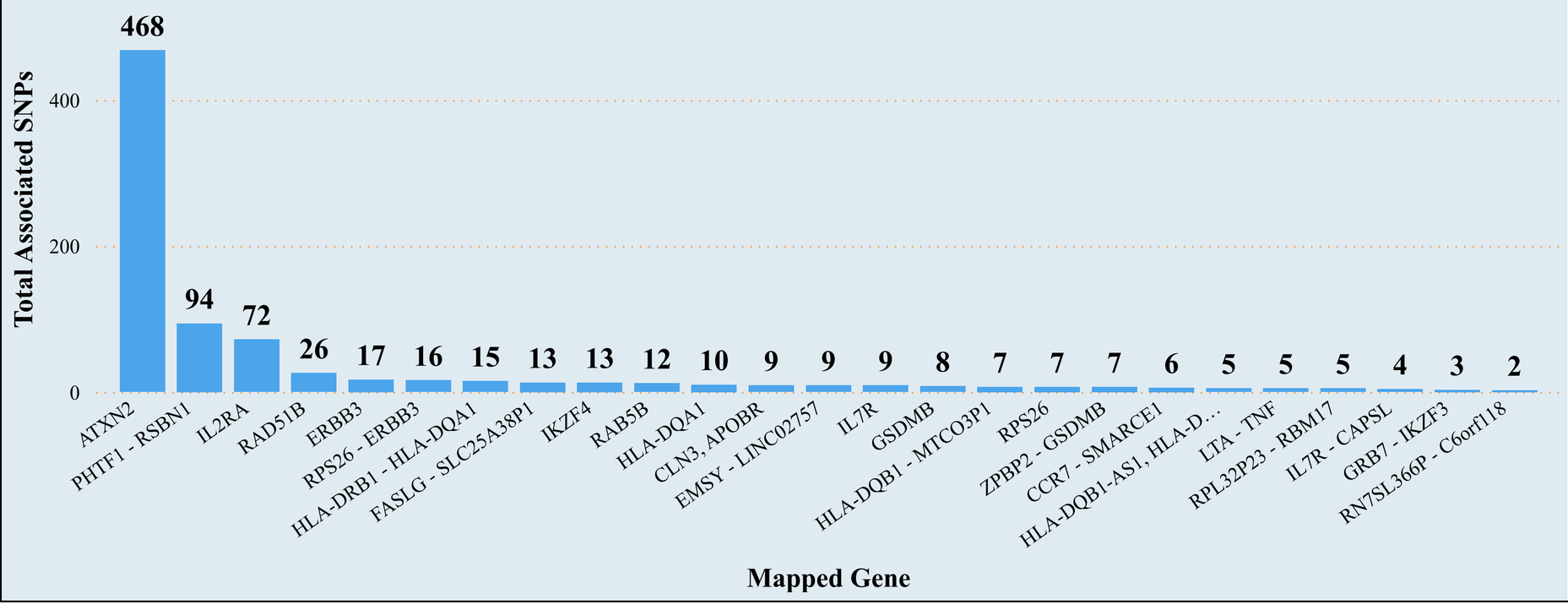
Top 5 Associated Diseases

Based on 30 shared SNPs across full GWAS catalog



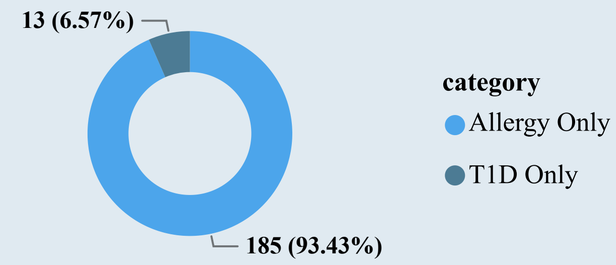
Top Genes Mapped to Shared T1D-Allergy SNPs

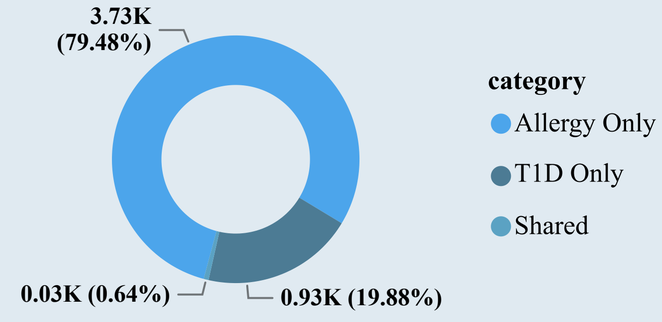
ATXN2 and immune-related genes dominate shared loci



Black Populations Only

198 SNPs total — no shared T1D-Allergy loci found



All populations

4,687 SNPs total across full GWAS catalog